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Original Article

Strategies for Opioid Minimization Following Total Knee Arthroplasty: A Comprehensive Review

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ABSTRACT

The ongoing opioid epidemic has prompted a reexamination of perioperative pain management, especially in total knee arthroplasty (TKA)—a procedure known for its high amount of postoperative pain and historical reliance on opioids. Among strategies for opioid-naïve patients, three broad approaches have emerged: Quantity limitation, dynamic reassessment-based prescribing, and tiered, multimodal pain regimens. While limiting prescription size and scheduling timely follow-ups remain important tools, perhaps an important approach to consider is a tiered, multimodal pain management regimen. This strategy begins with baseline administration of non-opioid agents such as acetaminophen, non-steroidal anti-inflammatory drugs (NSAIDs), and gabapentinoids, escalating only as needed to tramadol and, if necessary, stronger opioids. Preoperative cryoneurolysis, intraoperative regional nerve blocks, and long-acting local anesthetics further enhance this regimen's ability to minimize opioid exposure. These clinical gains are now reinforced by the Non-Opioids Prevent Addiction in the Nation (NOPAIN) Act, which provides separate Medicare reimbursement for select non-opioid pain treatments beginning in 2025, helping to eliminate financial barriers to adoption of these measures. In addition, real-world data—including results from the Innovations in Genicular Outcomes Research (iGOR) registry—have demonstrated the effectiveness of these techniques in reducing opioid use and improving functional and quality-of-life outcomes following TKA. Together, this convergence of clinical strategy, supportive policy, and data infrastructure provides a scalable and sustainable framework for advancing opioid stewardship in orthopaedic surgery without compromising patient comfort or recovery.

Keywords perioperative pain management, total knee arthroplasty, multimodal anesthesia, non-operative management, cryoneurolysis, multicenter registry

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The Opioid Crisis: A National Emergency

The opioid epidemic is a major public health crisis in the United States. Since 1999, the rate of opioid overdose deaths has tripled, with over 75,000 opioid-related fatalities reported in 2022 alone.[1] The most commonly implicated agents—oxycodone and hydrocodone—account for a major share of both prescription use and overdose deaths.[2] More than 10 million people in the United States inappropriately used prescription opioids in 2021, and an estimated 2.7 million Americans live with an opioid use disorder.[1] Economically, the opioid crisis exacts a toll exceeding $78.5 billion annually, encompassing health care expenditures, addiction treatment costs, lost productivity, and criminal justice involvement.[3] Beyond the morbidity, mortality, and economic burden, the opioid crisis has had a major societal impact: Children are orphaned, families are destabilized, and rural hospitals and emergency departments are overwhelmed.[4] The accessibility of high-potency synthetic opioids such as fentanyl, now a leading driver of overdose deaths, has only intensified the crisis.[5]

In response to these events, many efforts have emerged to curtail unnecessary opioid use and implement safer pain management practices.[6] Notably, surgical care is a major contributor[7]: Studies have shown that up to 10% of previously opioid-naïve surgical patients develop new chronic opioid use, often originating from routine perioperative prescribing.[2,8] In addition, increasing opioid use in the postoperative period following total joint arthroplasty has been associated with increased risk of venous thromboembolism and periprosthetic joint infection.[9] A retrospective study by Howard et al with 2,392 surgical patients found that the quantity of opioid prescriptions was associated with higher opioid consumption and increased patient-reported pain levels.[10,11] Thus, rethinking pain management in surgical settings—especially in painful operations like total knee arthroplasty (TKA)—has become a public health imperative.

Pain Following Total Knee Arthroplasty: A High-Stakes Surgical Challenge

Among elective orthopaedic procedures, TKA is recognized as one of the most painful.[12] Jane E. Brody from The New York Times famously described TKA as being "second only to childbirth" in

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intensity of postoperative pain, echoing the sentiments of countless patients and providers.[13] The pain is multifactorial—originating from extensive bone cuts, soft tissue releases, and manipulation of the joint capsule and neurovascular structures.[14] Without proper management, pain can delay rehabilitation, prolong hospital stays, and increase the risk of chronic pain syndromes.[4] Patients undergoing TKA fall broadly into two categories: Opioid-naïve (those who have little to no prior opioid exposure) and opioid-tolerant or dependent individuals.[14,15] Each group requires individualized perioperative strategies. Opioid-naïve patients are particularly vulnerable to postoperative exposure leading to persistent use, while opioid-tolerant patients often require escalated dosing with diminished analgesic efficacy.[16] Current estimates suggest that between 5% and 10% of patients undergoing TKA present on chronic opioids preoperatively—a subgroup with markedly higher risk for complications, poor outcomes, and long-term dependencies.[17] A retrospective study of 66,950 patients undergoing TKA found that 5.0% of the opioid-naïve cohort became chronic opioid users postoperatively relative to 34.8% who were taking opioids preoperatively.[18] To better educate patients on postoperative pain management, especially those who are opioid-dependent, a pain management agreement outlining the expectations and responsibilities for both the patient and surgeon. The agreement should focus on minimizing opioid use to manage pain effectively. By clearly defining expectations and consequences, the agreement can help prevent opioid misuse and encourage the use of multimodal strategies, leading to better pain control.

Current Perioperative Pain Management Strategies in Total Knee Arthroplasty

Modern pain management in TKA has shifted toward multimodal analgesia (MMA)—a strategy that targets various pain pathways using complementary, non-opioid agents (**Table 1**).[14] This approach not only improves analgesic efficacy but also reduces the need for opioids. Preemptive analgesia is beneficial and should be included in an MMA protocol. Preoperatively, enhanced recovery protocols often include acetaminophen, non-steroidal anti-inflammatory drugs (NSAIDs, like celecoxib), gabapentinoids (e.g., gabapentin or pregabalin), and even antidepressants like nortriptyline to reduce central sensitization.[19,20]

Intraoperatively, regional anesthesia is increasingly utilized. Techniques such as adductor canal blocks (ACBs), interspace between popliteal artery and capsule of the knee (iPACK) blocks, and periarticular local infiltration with anesthetic agents like liposomal bupivacaine (Exparel, Pacira BioSciences, Inc., Tampa, FL) are now standard in many institutions.[17] While regional anesthesia and nerve blocks have been widely used for postoperative pain relief, the optimal approach is still debated. An ACB and periarticular injection (PAI) provide postoperative analgesia following TKA without impacting quadriceps muscle function. The combined approach of ACB and PAI has been reported to reduce postoperative narcotic utilization. However, there is a range of techniques, variations in administration, and types of medications used with ACB. There are also numerous agents currently used in PAI cocktails, including local anesthetics, NSAIDs, corticosteroids,

Table 1 Multimodal analgesia options for total knee arthroplasty

Phase	Intervention	Purpose
Preoperative	Acetaminophen (1,000 mg PO)	Preemptive analgesia and reduce central sensitization
	NSAIDs/COX-2 inhibitors (e.g., celecoxib 200 mg)	Reduce inflammatory pain and peripheral sensitization
	Gabapentinoids (gabapentin 300–600 mg, pregabalin 75 mg)	Control neuropathic pain, reduce opioid need
	Nortriptyline (25 mg at night)	Enhance sleep, reduce central pain sensitization
	Cryoneurolysis (Iovera)	Reduce afferent nociceptive input and opioid need
Intraoperative	Adductor canal block	Regional block for anterior medial knee pain
	iPACK block	Regional block for posterior capsule pain
	Periarticular injection (e.g., Exparel, Zynrelef)	Local prolonged anesthesia and anti-inflammatory action
	Intravenous lidocaine or ketamine	Reduce opioid need, anti-hyperalgesic effects
	Neuraxial anesthesia (e.g., spinal with intrathecal morphine)	Reduce pain when PNB not feasible or as adjunct
Postoperative	Acetaminophen (650–1,000 mg PO every 6–8 hours)	Baseline analgesia
	NSAIDs (e.g., meloxicam, celecoxib)	Sustained anti-inflammatory effect
	Nortriptyline (10–25 mg at bedtime), suzetrigine (100 mg orally once then 50 mg twice a day)	Neuropathic pain control, non-opioid
	Cyclobenzaprine (5–10 mg thrice a day)	Muscle relaxant
	Gabapentinoids (gabapentin, pregabalin)	Neuropathic pain control, opioid-sparing
	Tramadol (50–100 mg PO every 6 hours PRN)	Breakthrough pain control (lower abuse profile)
	Opioids (as last resort—e.g., codeine and morphine then hydrocodone and oxycodone if needed)	Escalation only if other methods fail
	Non-pharmacologic (TENS, PT, lidocaine patches)	Enhance recovery, reduce narcotics, multimodal synergy

Abbreviations: h, hours; iPACK, interspace between popliteal artery and capsule of the knee; NSAID, non-steroidal anti-inflammatory drug; PO, orally; PT, physical therapy; TENS, transcutaneous electrical nerve stimulation; PRN, pro re nata.

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and sympathetic nervous system modulators. The two most common local anesthetics are bupivacaine and ropivacaine. The interest in liposomal bupivacaine has increased because of its prolonged duration of action compared with traditional bupivacaine. Non-anesthetic analgesics, including NSAIDs, such as ketorolac, are commonly used in PAIs to reduce the inflammatory process postoperatively. Systemic nervous system modulators such as epinephrine and clonidine are additional adjuncts to the PAI. Both of these drugs cause local vasoconstriction and decrease the absorption of the peripherally administered medications. Despite institutional variation, these techniques provide sustained pain relief, reduce systemic side effects, and facilitate same-day or next-day discharge. In some centers, cryoneurolysis of the superficial/deep genicular nerves using Iovera is implemented preoperatively to reduce pain signaling after TKA (Iovera, Pacira BioSciences, Inc., Tampa, FL).[21]

Postoperatively, MMA protocols typically rely on scheduled acetaminophen, NSAIDs, and gabapentin, progressing only to tramadol, codeine, or morphine in a stepwise manner for breakthrough pain. Adjunctive therapies like lidocaine patches, transcutaneous electrical nerve stimulation (TENS) units, and physical therapy further contribute to minimizing narcotic needs.[20,21] As evidence continues to increase, numerous centers now aim for opioid-free pathways, especially in opioid-naïve patients, with studies showing improved outcomes and higher patient satisfaction scores.

Perioperative Strategies for Opioid-Naïve Patients: Three Different Approaches

By promoting clear communication and shared responsibility with an appropriate MMA protocol, opioid use can be limited, and given the known risk that opioid-naïve patients may transition to persistent opioid use after surgery, targeted prescribing strategies have emerged to address this vulnerable group. These can be broadly divided into three approaches by the practicing orthopaedic surgeon[21]:

1. Quantity limitation
2. Dynamic prescription
3. Tiered, multimodal pain management regimen

The first foundational approach is quantity limitation—simply prescribing fewer opioids postoperatively. This "less is more" model is increasingly supported by real-world data showing that many patients consume only a fraction of what is prescribed. Studies have shown that when providers reduce the default prescription size from 60 to 10–15 pills,[10,11] pain scores remain stable and refill requests do not markedly increase. This reflects both changing provider behavior and improved patient education on expectations after TKA.

Another effective strategy emphasizes continuous reassessment and dynamic prescribing, as exemplified in the practice of one of us (M.A.M.). As a practicing orthopaedic surgeon in Baltimore, this was developed as a postoperative pain protocol adherent to the Maryland One-Week Rule,[22,23] a regulation enacted to curb opioid overprescription and reduce the risk of inappropriate usage. Under this rule, prescribers are encouraged—if not required in certain contexts—to limit initial opioid prescriptions for acute pain to a duration of no more than 7 days, unless a documented, clinically justifiable reason exists for a longer course.[23] The law aims to ensure that patients are not automatically given large quantities of opioids without proper follow-up and reassessment. In accordance with this rule, typical prescriptions are a conservative initial quantity of 10 to 20 pills, tailored to the patient's expected postoperative needs. Rather than automatically renewing the prescription, weekly evaluations to assess the patient's pain level, functional status, and overall recovery are performed.[21] If the patient continues to experience meaningful pain that interferes with recovery, a short-term refill is provided; otherwise, the opioid therapy is discontinued.[2] This model enforces regular patient monitoring and facilitates timely intervention if complications arise, while simultaneously minimizing the potential for opioid diversion, misuse, or the accumulation of unused pills in the community.

The third and increasingly influential approach involves the use of a tiered, multimodal pain management regimen, as implemented by another one of us (V.D.).[21] This tiered, multimodal pain management regimen often circumvents opioid use entirely, and the published results on a cohort of 144 patients undergoing TKA demonstrated significant reductions in opioid use at 3 months (55.6% opioid-free vs. 4.3%, $p < 0.0001$).[21] This protocol, based on evidence-based pharmacology, begins with the scheduled administration of acetaminophen and NSAIDs, followed by gabapentin and nortriptyline for neuropathic modulation. In cases of moderate to severe pain, tramadol is used judiciously, with true opioids reserved as a last-tier intervention. Importantly, this protocol does not rely on reflexive opioid prescribing but tailors escalation to actual patient needs, preserving efficacy while reducing risk.

Preoperative and Intraoperative Optimization

One of the senior authors (V.D.) has a pain management protocol that is anchored in a proactive, multimodal framework that begins well before the first incision (**Table 2**). The preoperative strategy aims to reduce the patient's baseline pain sensitivity and minimize the anesthetic and narcotic burden intraoperatively.[21] Patients are started on scheduled acetaminophen (500 mg every 6 hours), celecoxib (200 mg twice a day), and pregabalin (75 mg twice a day) preoperatively.

One unique component of this protocol is the use of Iovera (Pacira BioSciences, Inc., Tampa, FL),[24–26] a cryoneurolysis device that delivers cold therapy to sensory nerves such as the infrapatellar branch of the saphenous nerve. Applied preoperatively within 1 to 2 weeks prior to TKA, Iovera has been shown to reduce postoperative opioid use and improve early functional outcomes and now supported by separate Medicare reimbursement under the Centers for Medicare & Medicaid Services (CMS) outpatient payment policy effective 2025.[27] During surgery, a combination of regional nerve blocks—including adductor canal and iPACK blocks—is administered, along with periarticular infiltration using long-acting local anesthetics such as Exparel or Zynrelef, both of which are also covered under the Non-Opioids Prevent Addiction in the Nation (NOPAIN) Act.[21,22] These intraoperative measures ensure patients wake with low pain levels, often without the need for systemic opioids.

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Table 2 Louisiana State University orthopaedics postoperative pain Rx protocol

Healthy, opioid-naïve surgical pain Rx protocol (if able to tolerate po)
• Mild pain expected: #1, #2
• Moderate pain expected: #1, #2, #3
Severe pain expected: #1, #2, #3, #4, escalate as needed
• If **chronic opioid user**, continue baseline narcotics
1. NSAIDs
• Celecoxib 200 mg BID
• No bleeding or bone healing impact
• Consider diclofenac, Ketorolac, ibuprofen... if unable to obtain
▪ Slight bleeding/bone healing risk, maybe lowered if TXA used
2. Acetaminophen
• Start 500 mg q6h
• Increase to 500 mg q4h, if needed
3. Suzetrigine
• 100 mg loading dose then 50 mg BID × 2 weeks
4. Gabapentin
• Start 300 mg q6h (1,200 mg/day)
• Increase up to 3,600 mg/day if needed
• Be careful in elderly (>80-year-old), start 900 mg/day
5. Nortriptyline (if no hx of arrhythmia or MI)
• Na channel blocker—reduces pain
• Mildly sedating (helps with sleep)
• 25 mg QHS
If cardiac risk factor, use Desipramine
• 25 mg QHS
• If side effects (drowsiness, dry mouth, gait disturbance), contact psych or discontinue
6. Tramadol
• Selective MU agonist, fewer side effects and abuse potential
• Start 50 mg q8h as needed
• Increase up to 100 mg q6h as needed
7. Codeine
• 30 mg q6h as needed
• Increase up to max 60 mg q6h as needed
8. Morphine
• If already on acetaminophen
• Start 5 mg q6h as needed severe/breakthrough pain
• Increase up to 15 mg q6h
• Start 5 mg q12h for severe/breakthrough pain as needed

Abbreviations: BID, twice a day; NSAID, non-steroidal anti-inflammatory drug; q4h, every 4 hours; q6h, every 6 hours; q12h, every 12 hours; TXA, tranexamic acid; QHS, every night at bedtime; MI, myocardial infarction; MU, Mu opioid.

Postoperative Multimodal Management: A Tiered, Patient-Centered Regimen

Postoperatively, van Deventer et al employ a tiered pain management algorithm based on expected pain severity and individualized patient response.[21] For most opioid-naïve patients, analgesia is achieved using a combination of acetaminophen and NSAIDs alone. Gabapentin and cyclobenzaprine are added next if the patient contacts the office for additional pain management. More recently, Suzetrigine has been found to significantly improve acute pain and is now part of the multimodal regimen based on Dasa's early clinical results. A tricyclic antidepressant, such as nortriptyline (25 mg once daily at nighttime), is added next if needed, for patients who do not have cardiac contraindications to target central sensitization and improve sleep. This regimen is designed to address the patient's pain control in a deliberate and stepwise manner with non-opioid agents and improve tolerance to surgical insult.[19] If additional pain control is needed, the next tier includes tramadol (50–100 mg every 6 hours as needed)—a selective μ-opioid receptor agonist with a lower abuse profile.[20] Only if these measures prove insufficient are true opioids (e.g., codeine or morphine) introduced, and in tightly controlled quantities.

This conservative, yet effective, approach has demonstrated impressive results: In a retrospective cohort study, van Deventer et al showed that 72% of opioid-naïve patients undergoing TKA remained opioid-free at 3 months when managed under this protocol, compared with only 5.4% in a traditional prescribing group.[21] Importantly, these patients reported comparable or improved functional outcomes and satisfaction scores. The protocol also incorporates non-pharmacologic adjuncts such as lidocaine patches, TENS therapy, and a heavy emphasis on early mobilization, reinforcing the multidimensional nature of pain management. At a systems level, this strategy aligns with evolving reimbursement incentives and regulatory mandates, offering a scalable model for institutions seeking to reduce opioid exposure without compromising care. Ultimately, we recommend that perioperative pain management should be tailored by individual practitioners to their practice and patient populations.

Managing the Opioid-Dependent Patient: Balancing Comfort and Stewardship

While most patients undergoing TKA are opioid-naïve, an estimated 5% to 10% present with preexisting opioid use, making them tremendously more challenging to manage perioperatively.[2,17] These patients are at higher risk of prolonged hospital stays, increased complications, and persistent postoperative opioid dependence.[12] The goals in this population differ: Rather than aiming for complete opioid avoidance, clinicians seek to prevent escalation, optimize non-opioid adjuncts, and ideally begin tapering narcotic use in the postacute setting. Tolerance to opioids often reduces the effectiveness of systemic analgesics, and withdrawal concerns can complicate anesthesia and recovery.[7]

Protocols for opioid-tolerant patients often involve continuing the patient's baseline opioid regimen perioperatively to avoid withdrawal symptoms while layering on a multimodal backbone consisting of gabapentin, NSAIDs, acetaminophen, and potentially ketamine or dexmedetomidine infusions during the intraoperative and immediate postoperative period.[2] As mentioned above, creating guidelines with a pain management agreement in this patient population should clearly define expectations and consequences to ensure safe and effective postoperative pain control. Enhanced communication with pain management teams, primary care providers, and behavioral health is crucial. Long-term success in this population requires integrated care pathways, structured

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tapering protocols, and—critically—patient buy-in. While opioid-free recovery may not be a realistic goal for these individuals, meaningful opioid minimization with functional recovery is both achievable and beneficial.

Policy Innovation: The NOPAIN Act and a New Era for Surgical Pain Management

Recognizing the central role of perioperative prescribing in the broader opioid crisis, Congress passed the NOPAIN Act in December 2022.[28,29] The legislation represents a major policy shift designed to help remove potential financial barriers to the use of non-opioid pain management alternatives for Medicare beneficiaries undergoing outpatient procedures. Effective January 1, 2025, the NOPAIN Act mandates separate reimbursement under the Hospital Outpatient Prospective Payment System (OPPS) and the Ambulatory Surgical Center (ASC) Payment System for certain Food and Drug Administration (FDA)-approved non-opioid drugs, biologics, and devices used to treat postsurgical pain.[25,30,31] The 11 products associated with the No Pain Act include liposomal bupivacaine (Exparel), bupivacaine and meloxicam extended-release (Zynrelef), bupivacaine solution (Posimir), bupivacaine HCl (Marcaine), lidocaine HCl (Xylocaine), cryoneurolysis system (Iovera), elastomeric pain pump (ClariFusion), bupivacaine and meloxicam (HTX-011/Zynrelef), diclofenac potassium oral solution (Cambia), ketorolac injection (Toradol), and acetaminophen IV (Ofirmev).

Among these, the NOPAIN Act specifically includes liposomal bupivacaine (Exparel) and cryoneurolysis (Iovera), both of which have demonstrated efficacy in reducing opioid consumption following orthopaedic procedures, including total knee arthroplasty. Recent data from the Innovations in Genicular Outcomes Research (iGOR)—a prospective, multicenter registry tracking patient-reported outcomes across diverse institutions—support the clinical utility of these interventions. In a multicenter registry study of opioid-naïve patients undergoing TKA, cryoneurolysis of superficial/deep genicular nerves (Iovera) was associated with a 72% reduction in postoperative opioid use compared with controls ($p < 0.001$), significantly lower pain scores ($p = 0.046$), and reduced sleep disturbance over a 6-month period.[25,31] In a comparative registry analysis across six non-surgical knee osteoarthritis treatments, triamcinolone acetonide extended-release (TA-ER) demonstrated the lowest pain scores and greatest functional improvement, with the highest rate of patients achieving the minimally clinically important difference (MCID) in both pain ($p < 0.001$) and function ($p < 0.003$).[31,32] Additional sleep analysis showed that 63% of TA-ER patients had improved sleep quality, making them twice as likely to achieve MCID for sleep disturbance compared with other groups ($p < 0.05$).[26,32] Also, in a cohort of 505 patients, TA-ER (Zilretta) was the only treatment associated with a statistically significant reduction in conversion to TKA within 6 months ($p = 0.002$).[26]

Crucially, the law prohibits the bundling of treatments into procedural payments or reducing the procedural reimbursement to offset the cost of non-opioid therapies.[29] This payment separation, while temporary from 2025 to 2027, opens a pathway for the widespread adoption of multimodal regimens that were previously financially disincentivized.

Data Infrastructure and Real-World Implementation: iGOR, Real-World Administrative Databases, and Industry Collaboration

With federal reimbursement now aligned to promote opioid-sparing innovation, real-world evidence is playing a pivotal role in shaping clinical adoption. The iGOR initiative aggregates outcomes data on non-opioid interventions across health systems, capturing patient-reported outcomes, narcotic use, and complications.[33,34] Early reports suggest substantial reductions in opioid prescriptions when cryoneurolysis (Iovera) or long-acting local anesthetics (liposomal bupivacaine, bupivacaine + meloxicam) are included as standard elements of perioperative protocols. As demonstrated in recent publications, the iGOR registry has provided compelling evidence that Iovera® reduced postoperative opioid use by 72% ($p < 0.001$), with Zilretta improving both pain ($p < 0.001$) and function ($p \leq 0.003$), respectively. Pacira BioSciences, Inc., manufacturer of both liposomal bupivacaine and cryoneurolysis, has actively sponsored randomized controlled trials (RCTs), economic modeling, and real-world effectiveness studies to support value-based care pathways for orthopaedic surgery.[25,26,31,32,34–37] These efforts have also informed payer policy, ensuring broader access across Medicare Advantage and commercial plans.

At the clinical level, institutions are beginning to implement tiered prescribing strategies based on surgeon preference and risk stratification, like those outlined earlier:

- Surgeon A follows a strict four-tier system, using true opioids only in extreme breakthrough cases.
- Surgeon B maintains low default pill counts with conditional refills.
- Others adopt tramadol-based regimens to avoid opioids altogether.

Despite these developments, there remain challenges. Patients discharged with inadequate instructions may seek opioids from outside sources, and leftover medications often remain in home medicine cabinets, resulting in potential diverted use by others. Addressing these system-wide gaps ultimately requires not only robust prescribing frameworks but also patient education, secure disposal programs, and interdisciplinary care coordination.[21,23]

Conclusion

The ongoing opioid epidemic demands a fundamental shift in how we manage perioperative pain, particularly in high-pain procedures like TKA. Fortunately, a growing body of evidence supports multimodal, opioid-sparing strategies that can effectively control pain while minimizing risk—especially in opioid-naïve patients. Apart from quantity limitation and dynamic prescription, perhaps the most favored approach currently is a tiered, multimodal pain management regimen. A tiered multimodal pain management regimen sequentially escalates analgesic therapies—starting with non-opioid medications like acetaminophen and NSAIDs, then adding agents like gabapentin, suzetrigine, tricyclics, or tramadol, and reserving opioids only for breakthrough pain. The implementation of the NOPAIN Act has further catalyzed the trend toward

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decreasing opioid use by removing financial barriers to non-opioid therapies and aligning reimbursement with evidence-based care. Combined with emerging data from real-world registries and claims databases, the field now has both the clinical tools and policy framework to drive meaningful change. The future of post-TKA pain management lies in personalized, multimodal protocols that prioritize function, safety, and opioid stewardship—an approach that positions orthopaedic surgeons not just as proceduralists, but as critical stewards of long-term patient well-being.

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Statements and Additional Information

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